

# Aman Gupta

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## Research Profile

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Computer Science graduate with an MSc in Advanced Computer Science (Distinction) and a research background in environmental IoT sensor analytics and machine learning. My MSc dissertation investigated the prediction of human movement near IoT devices. I am now extending that work through reproducible experiments focused on rare-event prediction, temporal evaluation and generalisation across sensing devices.

## Research Experience

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### IoT Human Movement Prediction — MSc Dissertation & Research Extension | 2021 – Present

- Investigated whether environmental IoT sensor measurements could be used to predict human movement near IoT devices as part of my MSc dissertation.
- Reconstructed and documented the original MSc methodology, dataset, modelling approach and results as a reproducible research record.
- Extended the analysis beyond the original work to examine random versus chronological evaluation, device-level generalisation, feature ablation and decision-threshold sensitivity.
- Evaluated Logistic Regression, Random Forest and Histogram Gradient Boosting using metrics including average precision, precision, recall and F1-score because movement events are extremely rare.
- Documented the dataset, codebook, methodology, experimental protocol, reproducibility requirements and research results in a public GitHub research repository.

## Research Interests

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Machine learning for IoT sensing; human movement prediction; sensor-data analytics; rare-event prediction; temporal generalisation; distribution shift; model evaluation; reproducible machine-learning workflows.

## Education

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### Birmingham City University, UK

Master of Science in Advanced Computer Science (Distinction)

January 2020 – June 2021

- Dissertation: IoT Analytics for Predicting Human Movement near IoT Devices

### Maharshi Dayanand University, India

Bachelor of Technology in Computer Science Engineering (First Division)

August 2011 – May 2015

## Technical Skills

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- **Programming & Data:** Python, SQL, MySQL, Pandas, NumPy, JavaScript, C#
- **Machine Learning:** Scikit-learn, Classification, Regression, Predictive Modelling, Model Evaluation, Imbalanced Data
- **Analytics & Visualisation:** Power BI, Microsoft Excel, Jupyter, Matplotlib, Exploratory Data Analysis
- **Tools & Platforms:** Git, GitHub, SQL Server, Macromatrix, Microsoft Office Suite

## Selected Projects

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- **Bike Sales SQL Analysis** — SQL joins, aggregation, revenue calculations, customer order analysis and relational sales data.
- **Amazon Prime Data Analysis** — Python, pandas and exploratory data analysis to examine dataset structure, patterns and trends.
- **Stock Market Portfolio Analysis** — Historical market data, returns, risk measures and portfolio calculations.
- **Data Science Coursework** — Python-based data analysis and machine-learning techniques applied to practical coursework datasets.
- **Opportunity Dataset Team Project** — Team-based data preparation, analysis, visualisation and reporting.

## Professional Experience

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### Concentrix — Advisor I, Customer Service | November 2025 – Present

- Review and resolve customer cases while following documented processes and service policies.
- Identify cases requiring escalation and maintain clear and accurate case records.
- Use structured problem-solving and communication skills to resolve customer issues efficiently.

**Team Member / Trainee Team Leader — KFC, Yum Brands** | *September 2022 – November 2024*

- Supported restaurant operations, customer service, stock management and team coordination in a high-volume environment.
- Progressed to Trainee Team Leader and supported day-to-day team supervision and operational activities.
- Used operational systems and procedures to support scheduling, inventory management and service delivery.
- Trained and supported new team members on operational procedures, software tools and safety requirements.

**Device Testing Volunteer — Birmingham City University** | *March 2021 – May 2021*

- Conducted benchmarking and usability tests on academic devices using PassMark.
- Provided feedback on hardware performance, compatibility and usability for academic environments.

**Research Portfolio**

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- GitHub: [github.com/kaashgarg19](https://github.com/kaashgarg19)
- Current research: [github.com/kaashgarg19/robust-iot-human-movement-prediction](https://github.com/kaashgarg19/robust-iot-human-movement-prediction)
- Original MSc repository: [github.com/kaashgarg19/iot-human-movement-prediction-msc](https://github.com/kaashgarg19/iot-human-movement-prediction-msc)
- ORCID: [orcid.org/0009-0007-4112-9993](https://orcid.org/0009-0007-4112-9993)